

Principle Component Analysis Algorithm (PCA) for Image Recognition

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Abstract— This paper aims mainly to recognize the important of algorithm computing to identify the facial image without human intervention. Life in the current era imposed on us to increase the level of security and speed in the search for information, and the most important information is the capability of recognizing and identifying a person by his face. Principle Component Analysis algorithm (PCA) is a useful statistical technique used for finding patterns in data of high dimension and that has found application in face recognition and image compression fields that are used for reduce dimension vector to better recognize images.

Keywords- Principle Component Analysis algorithm (PCA), image recognition, Image analysis, Face recognition

I. INTRODUCTION

The rapid evolve of information introduced new steps to increase the speed at which transactions are conducted daily and behaviors that we now cannot live without handled electronically. For example, smart cards or electronic passwords that allow people to access their account at any time, in any place around the world have replaced the paper work. But there is still a problem in this area, what would happen if someone has lost a bank card or forgotten password, certainly it will take time to solve this problem. Therefore, the development of

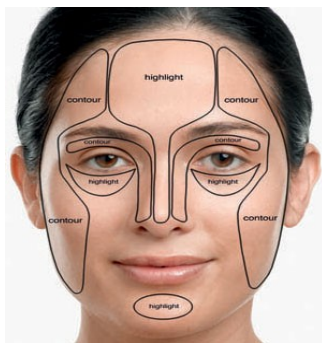


Figure 1. The regions that are either highlighted or contoured when used in PCA.[1]

mechanisms have been working on new measurements, like fingerprints, Iris, audio fingerprint and identify the user face. Facial recognition and identification features of modern technology will be the focus of this paper [1].

The automatic facial recognition is a relatively new concept that began in 1960. At first it was limited to determining the basic dimensions of the face - such as the eyes, the repositioning of nose and mouth. In the seventies of the last century, both Goldstein, Harmon and whiskey used 21 standards in the face to get to know the personal, such as the color of hair and thickness of lips. In 1988, both "Kirby" and "Serovic" apply the principle element analysis of the face, which was considered the most important step that led to the success of this technique later. Since then, studies and researches continued until this technique reached its current form [4].

Verification and identification of a person considered to be an important part on many levels. Face recognition techniques is used for personal security purposes. The importance of this technique is due to the fact that the modern facial recognition technology allows us to detect the facial expressions and facial changes in a perceivable range. It is even able to recognize the variation of local facial expressions of any individual. Nowadays face recognition is used widely by law forces to identify criminals who have been recognized in several crime scenes [4].

II. IMAGE PROCESSING STAGES

The face recognition algorithms based in comparing the face in the picture against all the faces in the device known in advance through the images and metadata, which is defined by the most important nodal points addressed by the software responsible which is the distance between the eyes and the nose area, width and depth of the cavity-kind and cheeks and place of their presence and jaw line and chin [5].



Figure 2. The three steps of image processing through PCA

First stage: We have two steps:

1. **Acquire:** this step is to get the picture and explore it. When any programs correspond to a weak resolution image will run a feature that helps to improve the image resolution and then the discovery of the faces in the picture.
2. **Detect:** At this step, where the face is determined and deduced from the big macro picture that was taken from the previous phase.

Second stage: We have four steps:

1. **Align:** It is the step where the alignment and standardization of the image (adjust the face angle with the camera angle).
2. **Extract:** is to extract basic features of the face of the image.
3. **Color update:** It is the step concerned with the color of the image and the extent of brightening the image.
4. **Compression:** pressing the image size to the maximum size possible so that it is not a change in image quality.

Third stage: We have three steps:

1. **Translate to Digital:** converts the image into digital data.
2. **Pattern:** to create a form represents a digital image to be handled.
3. **Save:** to keep beam data that reflects the image of the face.

III. WHAT IS PCA ?

PCA or principal component analysis algorithm is one of the most effective techniques that have been used in the section of image compression. PCA is a biometric technique it is classified as one of statistical methods in this area. The main goal in the PCA algorithm lies in reducing

the large dimensions of space data to the dimensions of the smallest spaces.

Karl Pearson, an important and famous mathematician, created PCA in 1901. PCA is used in creating predictive models such as face recognition and in exploratory data analysis system. PCA is considered to be one of the true eigenvector-based multivariate analyses; however, it is also believed that PCA is the most simplified method among them. PCA functions on reducing the dimensions of the transformed data. This is accomplished by using simple principle components that can give the user a picture of the requested data but in smaller shadow-like images. PCA can detect any change of a data by revealing the composition of this data. The number of original variables (face images) is more than or equal to the number

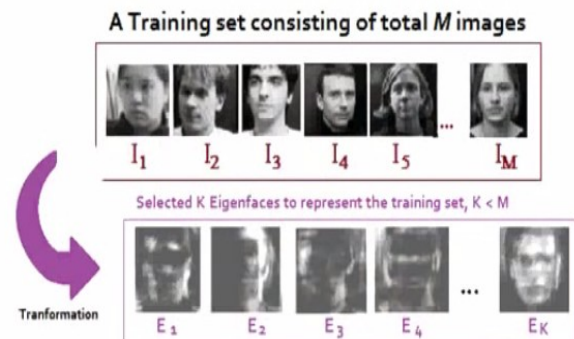


Figure 3. Shows the training set (M) images “top” and their compared Eigenfaces (K) “bottom”.

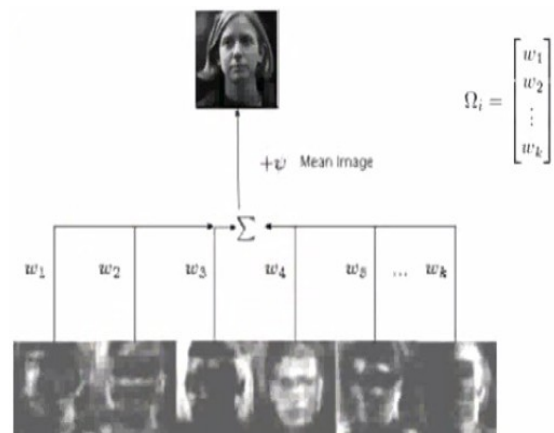


Figure 4. A Flow diagram shows 6 variations of Eigen faces that all represent the mean image above.

Of principal components (Eigen faces) $M \geq K$. Therefore, each variable in the original dataset can be represented in terms of these K principal components [5].

PCA can be done by two main procedures. The first is achieved by decomposition of the eigenvalue of a data covariance. On the other hand, the second method is done by decomposition of a singular value of a data matrix. PCA results are expressed as a component or factor scores (data points represents “how much” K principal component equals) and loadings (Standardized component score weight). Therefore, the resulted image can be expressed as the value of each K Eigen faces it was related to. After reducing the dimensionality of the training set, the face images that are generated are referred as the Eigen face (or eigenvectors). In the training set, each image is referred to one of the selected Eigen faces. The unknown faces are

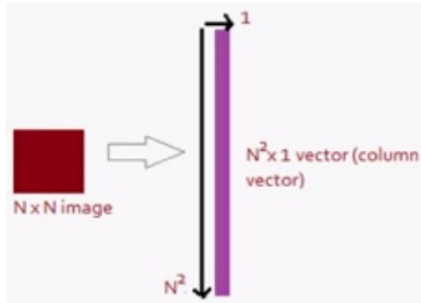


Figure 1. The distance between the image and the vector is calculated.

compared to the Eigen faces of the training set and the distance between both images is calculated [4].

PCA Eigenfaces method considers each pixel in an image as a separate dimension. So an $N \times N$ image has N^2 pixels, therefore has N^2 dimensions. For recognition, we need a training set (dataset) of face images. PCA Eigenfaces method does not work on images directly. It first converts them to matrix (vector) from. [4]

IV. STEPWISE OF PCA ALGORITHM[3]

1- Step 1: Create a training set and load it.



Figure 6. Step 1.

2- Step 2: Convert face images in Training set to face vectors.

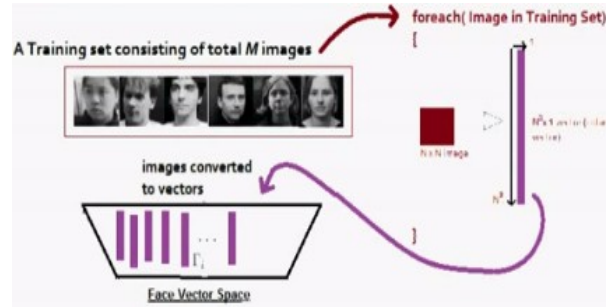


Figure 7. Step 2.

3- Step 3: Normalize the face vectors.

- calculate the average face vector
- subtract average face from each face vector $\phi = T_i - \Psi$ to calculate

Figure 6.4 eigenvectors we need to calculate the PCA step 5 covariance matrix C

$$C = AA^T \quad \text{where } A = \{\phi_1, \phi_2, \phi_3, \dots, \phi_M\}$$

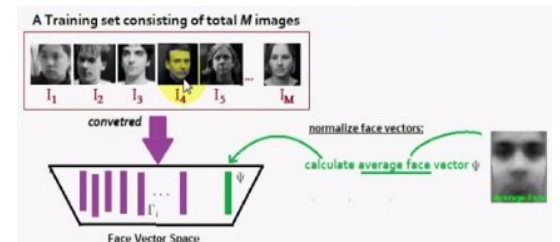


Figure 8. Step 3.

4- Step4: Dimensionality of the training set is decreased.

5- Step5: Eigenvectors are calculated from the covariance matrix.

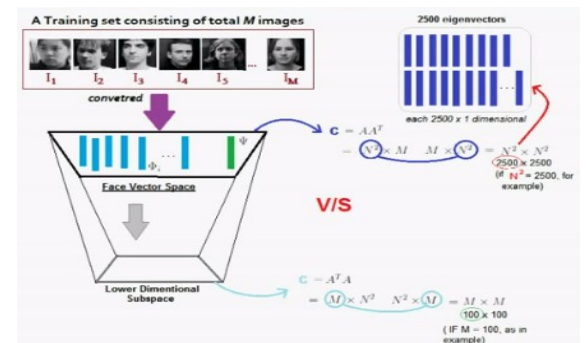


Figure 9. Step 4 and 5.

6- Step 6: Select k best eigenfaces, such that $K < M$ and can represent the whole training set.

Note: selected k eigenfaces must be in the original dimensionality of the face vector space.

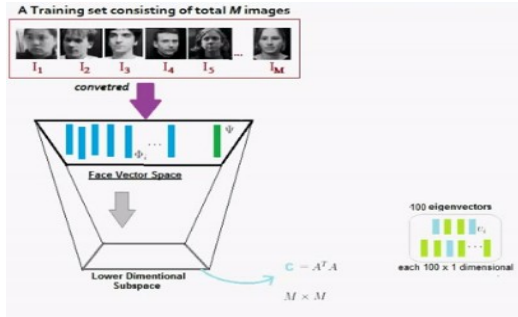


Figure 10. Step 6.

7- **step7:** Convert lower dimensional k eigenvectors to original face dimensionality

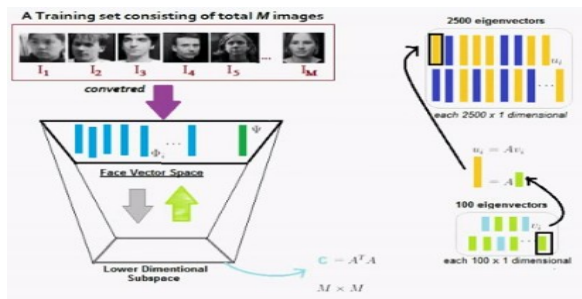


Figure 11. Step 7.

8- **Step 8:** Represent each face image in a linear combination of all K eigenvectors. A summation of the weight of the K Eigenfaces + the Mean face is calculated to refer to each face that is from the Training set.

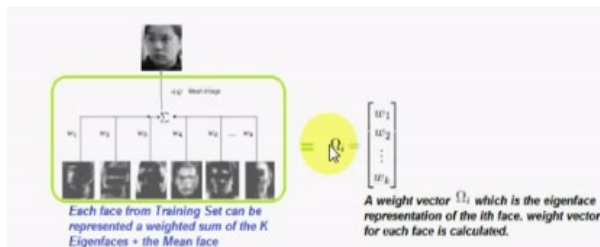


Figure 12. Step 8.

After that we can find out the eigenfaces for the M training images and selected a set of K most relevant Eigen faces, $K < M$. The related weight vectors are calculated and stored in the training set of each image individually.

V. THE OBJECTIVES OF PCA

PCA is a method that is able to describe and identify the face of a specific person. It has the following objectives:

- Producing a reduced number of orthogonal created variables (factors) that are achieved by reducing the original variables.
- Displaying connections between the original variables, and between the original variables and factor.
- Displaying correlations and similarities between statistical units.

PCA removes the correlations among the different input dimensions and significantly reduce the data dimensions. PCA is used to extract features of face images and construct the eigenface space.

VI. THE PROBLEM IN IMAGE RECOGNITION

There are many obstacles that prevent facial recognition , For Example [2]:

- A. If the lighting is directed to the face low or high.
- B. If the face is not directed to the camera completely.
- C. If the user moves parts of the face such as the eye or nose.
- D. If the user is wearing a hat or sunglasses.

VII. RESULTS AND DISCUSSION

When comparing the mean faces of the subjects that were used in our research with the Eigen faces of the training set a high facial detection ratio was showing. The result of the mean faces of the research subjects were likely similar to the Eigen faces of the training set. However, in some cases the results were not accurate due to some factors such as; low or high lighting towards the face, the face is not fully directed to the camera, or the subject was wearing a hat or sunglasses.

In general PCA showed an impressive recognition ability of different individuals with different face patterns. The result was highly positive and recommended. Further researches should be conducted to assist this research.

VIII. CONCLUSION AND FUTURE WORK

In this paper it is shown how it is read from the user's face and how they are handled within the programming environment. As demonstrated how facial recognition and how it is stored in memory and what is the mechanism that extrapolation of the faces of users has been pointed out in this paper algorithm property and how to deal with it and what its significance in science. In the outlook for this project it is recommended that the application of this algorithm in various sciences which contains many large and complex, interrelated data so as to analyze the data and access to low-and interrelated data.

IX. REFERENCES

- [1] Er M. J., Wu S., Lu J. and Toh H. L. "Face recognition with radial basis function (RBF) neural networks", IEEE Trans. Neural Networks, vol. 13, no. 3, pp. 697-710 2002
- [2] Yang M.-H., Ahuja N. and Kraegman D. "Face recognition using kernel Eigenfaces", Proc. IEEE Int'l. Conf. Image Proc., pp.37-40. 2000
- [3] J. Hightower, and G. Borriello "Principal component analysis", IEEE Computer and IEEE Special Report "Trends Technologies & Applications in Mobile Computing", August 2007
- [4] Analyzing PCA-based Face Recognition Algorithms: Eigenvector W. Zhao, R. Chellappa, A. Rosenfeld, and P. J. Phillips, "Face Recognition: A Literature Survey", ACM Computing Surveys, Technical Report, Vol. 35, pp. 399-458, Dec 2003
- [5] Ashok, J., Shivashankar, V., & Mudiraj, P. (2010). An overview of Biometrics. International Journal on Computer Science and Engineering, 2(7), 2402-2408.
- [6] Kekre, H., Thepade, S. D., & Chopra, T. (2010). Face and gender recognition using principal component analysis. *International Journal on Computer Science and Engineering*, 2(4), 959-964.